

Lecture 9: State Feedback & Observer

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
tillstånd	state
state-space (form/representation)	tillstånds(form/beskrivning)
Pole placement	Polplacering
State feedback	Tillståndsåterkoppling
steady state/equilibrium	stationärt tillstånd
observer	observatör
estimation error	skattningsfelet
measurement error	mätfelet
complementary sensitivity function	komplementära känslighetsfunktionen
sensitivity function	känslighetsfunktionen



Outline

Recap

State feedback

Controllable canonical form

External I control

Observer

Feedback from reconstructed states

Alternative implementation

In the previous lectures (F7, F8)..
► State-space model

$$\begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{cases} \quad x(0) = x_0$$

- Observability and Controllability (observable and controllable states)

Today:

- State feedback (from $x(t)$)
► State feedback with observer (from $y(t)$)



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State feedback

Assume that we feed back a system using $u = -Lx + l_0 r$.

Then, the closed-loop system will be of the form:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \\ u = -Lx + l_0 r \end{cases} \implies \begin{cases} \dot{x} = Ax + B(-Lx + l_0 r) \\ y = Cx \end{cases}$$

$$\implies \boxed{\begin{cases} \dot{x} = \underbrace{(A - BL)}_{A_c} x + \underbrace{B l_0}_{B_c} r \\ y = \underbrace{C}_{C_c} x \end{cases}}$$

It is described by the matrices $(A_c, B_c, C_c, D_c) := (A - BL, B l_0, C, 0)$.

$\implies$ The poles of the closed-loop system are given by the eigenvalues of $A - BL$, i.e., the roots of the equation $\det(sI - (A - BL)) = 0$.

Pole placement

We observe that

- ▶ $A - BL$ has n eigenvalues (= poles of closed-loop system)
- ▶ L has n parameters

If the system is *controllable*, the poles of the closed-loop system can be placed arbitrarily via appropriate selection of L [Resultat 9.1].

Suppose we have the system and the feedback law u (i.e., L and l_0) as follows

$$\begin{cases} \dot{x} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x \\ u &= -\begin{bmatrix} l_1 & l_2 \end{bmatrix} x + l_0 r \end{cases}$$

We calculate $A - BL$ as

$$A - BL = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} l_1 & l_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -l_1 & -l_2 \end{bmatrix}$$

Pole placement: Example

Example (cont'd)

The eigenvalues of $A - BL$ are given by

$$\det (\lambda I - (A - BL)) = \det \begin{bmatrix} \lambda & -1 \\ l_1 & \lambda + l_2 \end{bmatrix} = \lambda^2 + l_2\lambda + l_1$$

Now assume, for example, that we want to place the poles at -3 and -3 (double pole).

$$\Rightarrow (\lambda + 3)^2 = \lambda^2 + 6\lambda + 9$$

Comparing this with the equation above, we see that we should choose

$$\begin{cases} l_2 = 6 \\ l_1 = 9 \end{cases} \Rightarrow u = -9x_1 - 6x_2 + l_0 r$$

This gives us the closed-loop system

$$\begin{cases} \dot{x} &= \begin{bmatrix} 0 & 1 \\ -9 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} l_o r \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases}$$

If we had instead wanted to place the poles at -3000 and -3000 (approx. 1000 times faster), we would have needed to select

$$\Rightarrow \begin{cases} l_1 &= 9 \cdot 10^6 \\ l_2 &= 6 \cdot 10^3 \end{cases}$$

⇒ The choice of poles for the closed-loop system is limited by the size of $u(t)$!

How should one choose l_0 ?

Solution:

Choose l_0 so that $y(t) = r(t)$ in steady state when $r(t) = \text{constant}$, i.e. $G_c(0) = 1$.

Remember that $\dot{x} = 0$ in steady state (at equilibrium).

State feedback: Example

Example (cont'd)

We had the closed-loop system

$$\begin{cases} \dot{x} &= \begin{bmatrix} 0 & 1 \\ -9 & -6 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} l_0 r \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases}$$

At equilibrium and with a constant reference, we have

$$\begin{cases} r(t) &= \bar{r} \\ \dot{x} &= 0 \end{cases} \implies \begin{cases} \dot{x}_1 = x_2 = 0 \\ \dot{x}_2 = -9x_1 - 6x_2 + l_0 \bar{r} = 0 \end{cases}$$

which directly gives $x_2 = 0$ and $x_1 = \frac{l_0}{9} \bar{r}$.

However, since $y = x_1 = \frac{l_0}{9} \bar{r}$, we should choose $l_0 = 9$ so that $y = \bar{r}$.

$\implies$ state feedback law : $u = -9x_1 - 6x_2 + 9r$

The weakness of the method we used above to determine l_0 is that it requires knowledge of $G(0)$ and that no disturbances affect the system.

$\Rightarrow$ Introduce I control!

Summary of state feedback

The system:

$$\begin{cases} \dot{x} &= Ax + Bu \\ y &= Cx \end{cases} \Rightarrow G(s) = \frac{b_1 s^{n-1} + \dots + b_n}{s^n + a_1 s^{n-1} + \dots + a_n}$$

State feedback law: $u = -Lx + l_0 r$

Closed-loop system:

$$\begin{cases} \dot{x} &= (A - BL)x + Bl_0 r \\ y &= Cx \end{cases} \Rightarrow G_c(s) = \frac{(b_1 s^{n-1} + \dots + b_n)l_0}{s^n + \alpha_1 s^{n-1} + \dots + \alpha_n}$$

Poles of closed-loop system: $\det(sI - (A - BL)) = s^n + \alpha_1 s^{n-1} + \dots + \alpha_n = 0$

- We can influence the poles of the closed-loop system via L (optimization)
- Zeros remain unchanged

Controllable canonical form

Example (Controllable canonical form [Resultat 8.2]: Pole Placement [Exempel 9.2])

$$\dot{x} = \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_n \\ 1 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & \cdots & 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = [b_1 \ \cdots \ b_n]x$$

$$\Rightarrow A - BL = \begin{bmatrix} -\overbrace{(a_1 + l_1)}^{\alpha_1} & -\overbrace{(a_2 + l_2)}^{\alpha_2} & \cdots & -\overbrace{(a_n + l_n)}^{\alpha_n} \\ 1 & 0 & \cdots & 0 \\ \vdots & \ddots & 0 & 0 \\ 0 & \cdots & 1 & 0 \end{bmatrix}$$

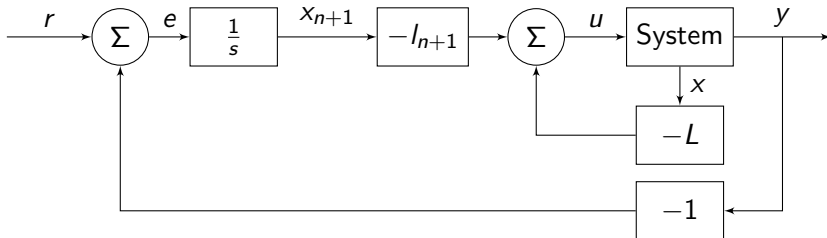


- Easy to determine L
- Easy to see that the zeroes are not affected (same $C = [b_1 \cdots b_n]!$)

$$\Rightarrow l_0 = \frac{\alpha_n}{b_n}$$

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External I control



We resort to a trick and introduce *extra states*

$$x_{n+1} = \int_0^t e(\tau) d\tau = \int_0^t (r(\tau) - y(\tau)) d\tau \Leftrightarrow \dot{x}_{n+1} = -Cx + r$$

.. that is, we extend the state-space model as follows:

$$\underbrace{\begin{bmatrix} \dot{x} \\ \dot{x}_{n+1} \end{bmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x \\ x_{n+1} \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{\mathbf{B}} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$$


$$u = -Lx - l_{n+1}x_{n+1} \quad \Leftrightarrow \quad u = -\mathbf{L}\mathbf{x}$$

At steady state, we have:

$$\begin{cases} \dot{x} &= 0 \\ \dot{x}_{n+1} &= r - y = 0 \end{cases}$$

despite, for example, step disturbances (stegstörningar) and model errors.



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What should one do if all state variables *cannot* be measured accurately?

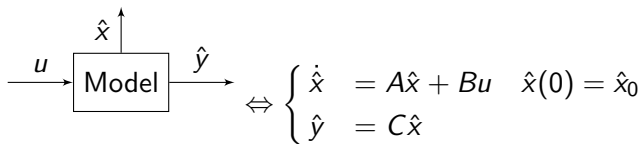
Answer: Introduce an *observer* (estimates the states)

For example, we can measure position and thereby estimate velocity.

Let's assume we have a model:

$$\begin{cases} \dot{x} &= Ax + Bu & x(0) = x_0 \\ y &= Cx \end{cases}$$

The idea is to simulate the system:



Error signal

$$y(t) - \hat{y}(t) = y(t) - C\hat{x}(t)$$

is a measure of how well the simulation works.

Use feedback! $\Rightarrow$ Observer (Kalman filter)

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + Bu + K(y - C\hat{x}) \\ \Leftrightarrow \dot{\hat{x}} &= (A - KC)\hat{x} + Bu + Ky\end{aligned}$$

Note that K is an $n \times 1$ vector (P control).



Observer: How do we choose K ?

The *estimation error* $\tilde{x}$ is given by:

$$\begin{aligned}\dot{\tilde{x}} &= \dot{x} - \dot{\hat{x}} = Ax + Bu - A\hat{x} - B\hat{u} - K(Cx - C\hat{x}) \\ &\implies \dot{\tilde{x}} = (A - KC)\tilde{x},\end{aligned}$$

where $\tilde{x}(0)$ is the initial error.

The solution to $\dot{\tilde{x}} = (A - KC)\tilde{x}$ is given by

$$\tilde{x}(t) = e^{(A-KC)t} \tilde{x}(0)$$

Note that:

- ▶ $\tilde{x} \rightarrow 0$ as $t \rightarrow \infty$ if the eigenvalues of $A - KC$ lie in LHP
- ▶ Using K , we can influence how quickly $\tilde{x} \rightarrow 0$
- ▶ If the system is *observable*, the **eigenvalues of $A - KC$ can be placed arbitrarily** [Resultat 9.2]

The *measurement error* is given by $y_m = y + e$.

$$\Rightarrow \dot{\tilde{x}} = (A - KC)\tilde{x} + Ke$$

Large $K \Rightarrow$ Sensitive to measurement errors

Small $K \Rightarrow$ Slow

Optimize $\Rightarrow$ Kalman filter!

Rule/Tumregel: Choose $|\text{eig}(A - BL)| \leq |\text{eig}(A - KC)|$

Consider the system

We calculate

and then

(poles of the closed-loop system in -3).

Example (cont'd)

Place the observer poles in, for example, -4 :

$$(s + 4)^2 = s^2 + 8s + 16$$

$$\Rightarrow \begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 16 \end{bmatrix}$$

Answer:

$$\dot{\hat{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \hat{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 8 \\ 16 \end{bmatrix} (y - [1 \ 0] \hat{x})$$



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System:

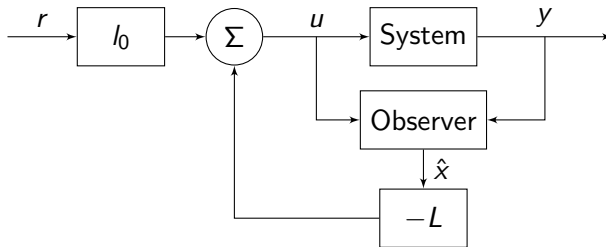
$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}$$

Observer:

$$\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})$$

Feedback law from the reconstructed states:

$$u = -L\hat{x} + l_0 r$$



Feedback from reconstructed states

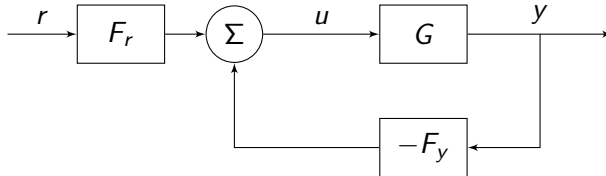
Closed-loop system? (Order $2n$)

$$G_c(s) = C(sI - (A - BL))^{-1}B l_0$$

Identical to G_c obtained using a state feedback law, because $\hat{x}(0) = x(0) = 0$ (follows using transfer function algebra/överföringsfunktions-algebra)

Clarification: $\tilde{x}$ is not controllable, but observable. The corresponding poles are truncated.

Alternative implementation



$$F_r = \left(1 - L(sI - (A - BL - KC))^{-1}B\right)l_0$$

$$F_y = L(sI - (A - BL - KC))^{-1}K$$

"Pole placement"

Complementary sensitivity function

The *complementary sensitivity function* is given by

$$T(s) = G_c(s) \times \underbrace{L(sI - (A - KC))^{-1}K}_{L, K, \text{ and the observer's dynamics affect robustness}}$$

The usual *sensitivity function* is given by

$$S = 1 - T$$



We are done with state-space theory..

Next time: Controller structures and IMC